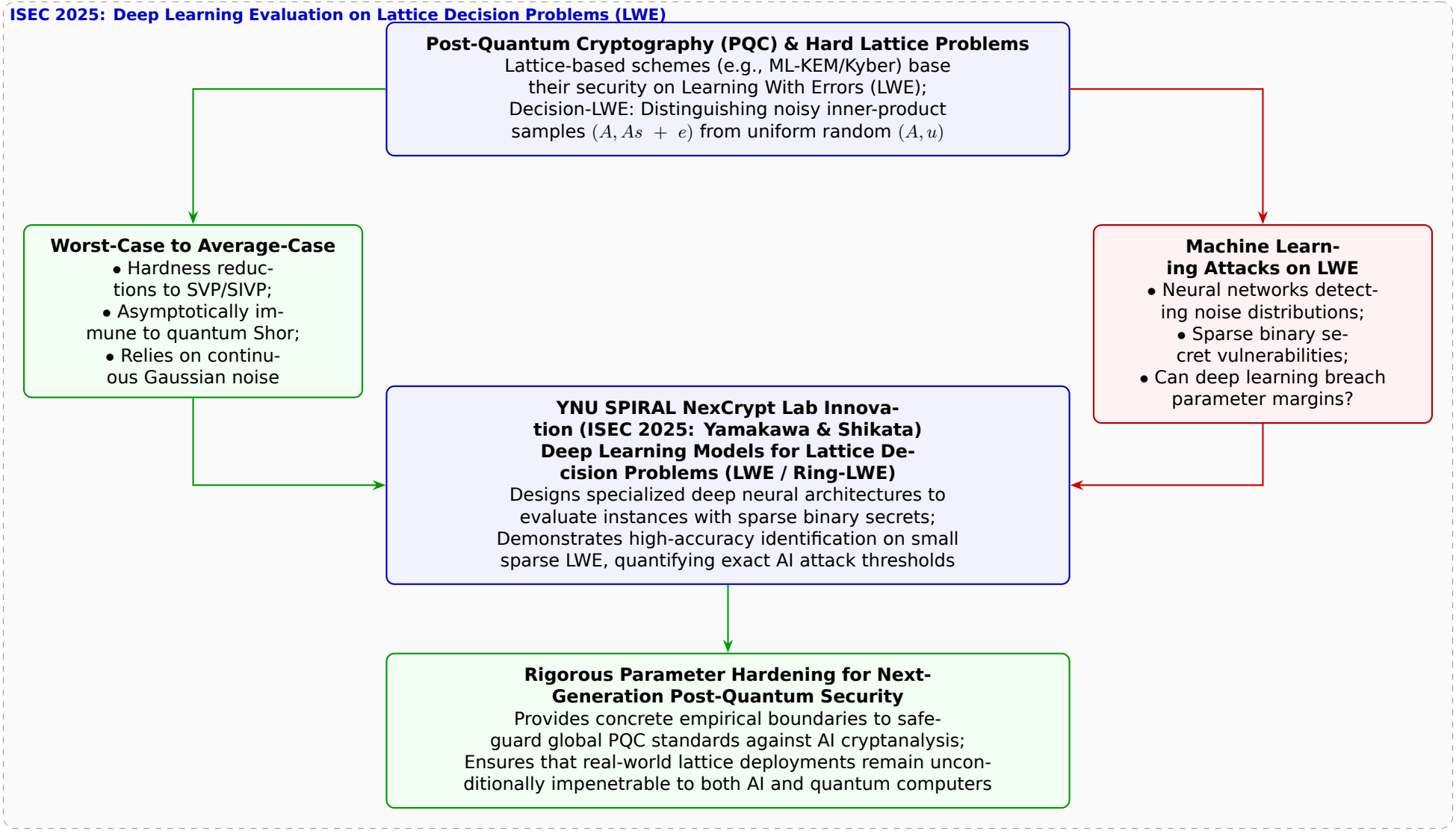

ISEC 2025: Deep Learning Evaluation on Lattice Decision Problems (LWE)
Post-Quantum Cryptography (PQC) & Hard Lattice Problems
Lattice-based schemes (e.g., ML-KEM/Kyber) base their security on Learning With Errors (LWE); Decision-LWE: Distinguishing noisy inner-product samples $(A, As + e)$ from uniform random $(A, u)$
Worst-Case to Average-Case
• Hardness reductions to SVP/SIVP;
• Asymptotically immune to quantum Shor;
• Relies on continuous Gaussian noise
Machine Learning Attacks on LWE
• Neural networks detecting noise distributions;
• Sparse binary secret vulnerabilities;
• Can deep learning breach parameter margins?
YNU SPIRAL NexCrypt Lab Innovation (ISEC 2025: Yamakawa & Shikata)
Deep Learning Models for Lattice Decision Problems (LWE / Ring-LWE)
Designs specialized deep neural architectures to evaluate instances with sparse binary secrets; Demonstrates high-accuracy identification on small sparse LWE, quantifying exact AI attack thresholds
Rigorous Parameter Hardening for Next-Generation Post-Quantum Security
Provides concrete empirical boundaries to safeguard global PQC standards against AI cryptanalysis; Ensures that real-world lattice deployments remain unconditionally impenetrable to both AI and quantum computers